

A. Collatz Conjecture

time limit per test: 1 second

memory limit per test: 256 megabytes

You are doing a research paper on the famous Collatz Conjecture. In your experiment, you start off with an integer x , and you do the following procedure k times:

- If x is even, divide x by 2.
- Otherwise, set x to $3 \cdot x + 1$.

For example, starting off with 21 and doing the procedure 5 times, you get $21 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 8 \rightarrow 4$.

After all k iterations, you are left with the final value of x . Unfortunately, you forgot the initial value. Please output any possible initial value of x .

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 400$). The description of the test cases follows.

The first line of each test case contains two integers k and x ($1 \leq k, x \leq 20$).

Output

For each test case, print any possible initial value on a new line. It can be shown that the answer always exists.

Example

input	Copy
3 1 4 1 5 5 4	
output	Copy
1 10 21	

Note

In the first test case, since 1 is odd, performing the procedure $k = 1$ times results in $1 \cdot 3 + 1 = 4$, so 1 is a valid output.

In the second test case, since 10 is even, performing the procedure $k = 1$ times results in $\frac{10}{2} = 5$, so 10 is a valid output.

The third test case is explained in the statement.

Codeforces Round 1047 (Div. 3)

Finished

Practice



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Start virtual contest

→ Submit?

Language: Python 3.13.2

Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

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Submission	Time	Verdict
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→ Problem tags

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Announcement (en)